



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16239
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2813.15
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10724606	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16239
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

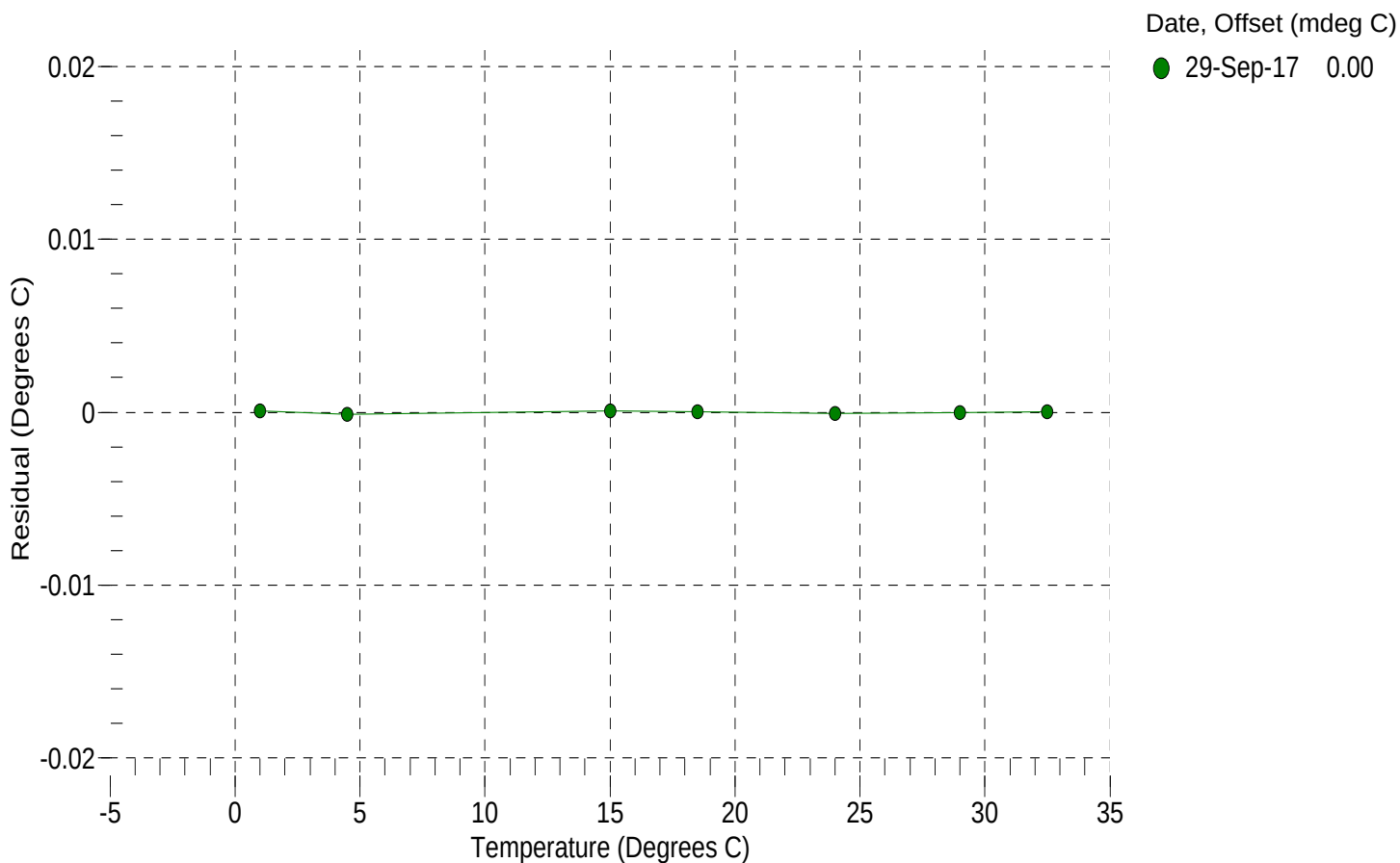
a0 = -1.529444e-004
a1 = 3.122891e-004
a2 = -4.674323e-006
a3 = 2.084168e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	565348.0	1.0001	0.0001
4.5000	484415.5	4.4999	-0.0001
15.0000	310708.7	15.0001	0.0001
18.5000	269619.2	18.5000	0.0000
24.0000	217029.5	23.9999	-0.0001
29.0000	179258.0	29.0000	-0.0000
32.5001	157319.6	32.5001	0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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CALIBRATION DATE: 29-Sep-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.965356e-001
h = 1.259214e-001
i = -5.457922e-005
j = 1.961003e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 2.0734e-007

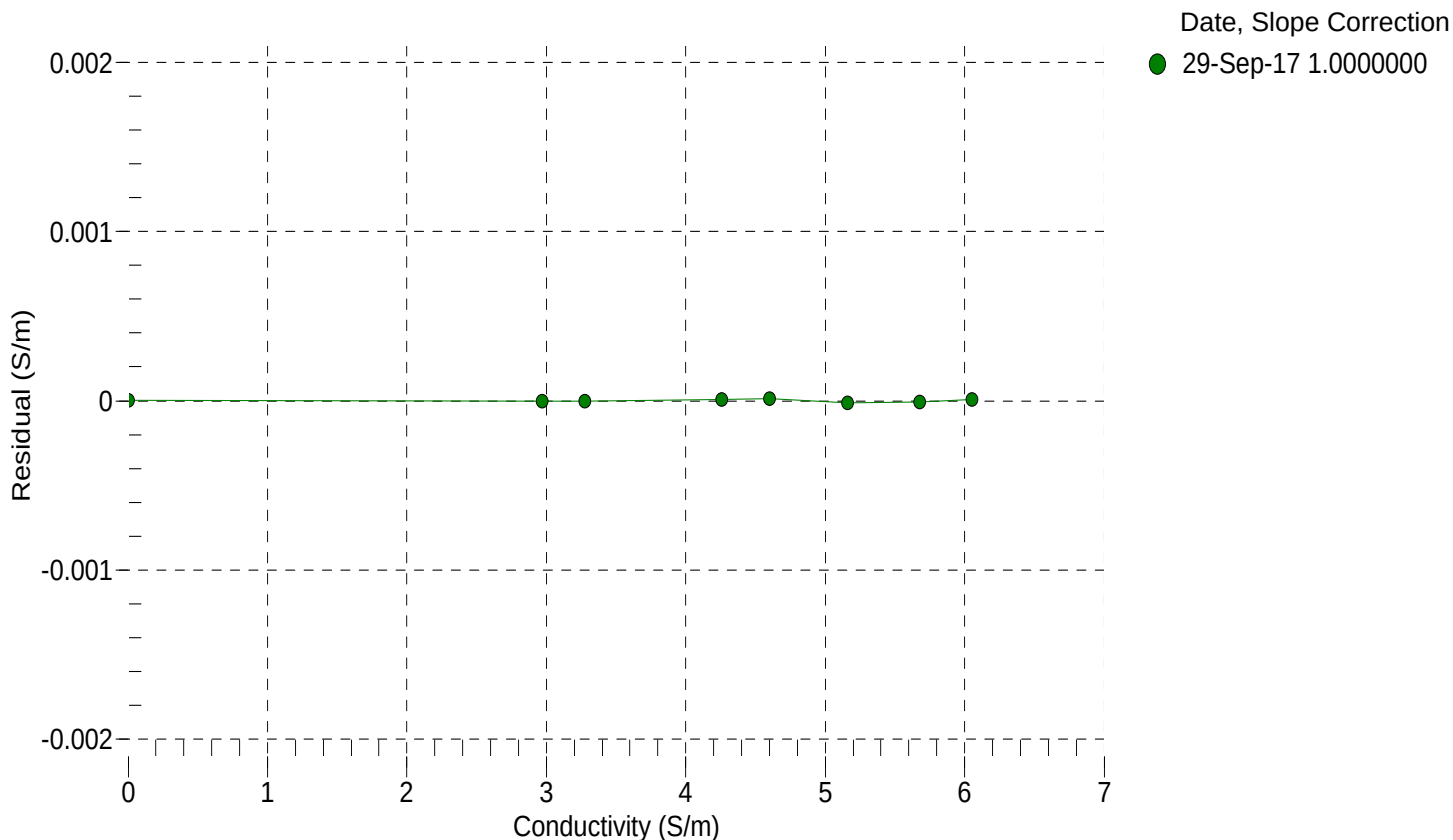
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2813.15	0.00000	0.00000
1.0000	34.7400	2.97006	5605.64	2.97006	-0.00000
4.5000	34.7203	3.27656	5817.38	3.27655	-0.00000
15.0000	34.6787	4.25652	6447.14	4.25652	0.00001
18.5000	34.6702	4.60107	6654.15	4.60108	0.00001
24.0000	34.6613	5.15811	6975.54	5.15810	-0.00001
29.0000	34.6566	5.67910	7262.98	5.67909	-0.00001
32.5001	34.6542	6.05092	7461.19	6.05093	0.00001

$f = \text{Instrument Output(Hz)} * \text{sqrt}(1.0 + \text{WBOTC} * t) / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

$\text{Conductivity (S/m)} = (g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16239
CALIBRATION DATE: 25-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10724606

COEFFICIENTS:

PA0 = -1.167255e-001
PA1 = 9.108054e-003
PA2 = -1.470889e-010
PTEMPA0 = -5.903056e+001
PTEMPA1 = 5.312767e-002
PTEMPA2 = -4.813755e-007

PTCA0 = 5.241275e+005
PTCA1 = 3.372427e+000
PTCA2 = -8.676218e-002
PTCB0 = 3.107923e+005
PTCB1 = 2.090306e+002
PTCB2 = -3.168466e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.65	525801.0	1511.0	14.69	0.00	32.50	1751	525839.50
614.89	592393.0	1518.0	614.83	-0.00	29.00	1683	525850.32
1214.84	659109.0	1521.0	1214.76	-0.00	24.00	1586	525858.32
1714.80	714821.0	1521.0	1714.77	-0.00	18.50	1479	525856.89
2314.71	781808.0	1523.0	2314.73	0.00	15.00	1412	525851.76
2914.62	848920.0	1523.0	2914.55	-0.00	4.50	1209	525836.70
2314.68	781815.0	1524.0	2314.76	0.00	1.00	1142	525828.49
1714.77	714834.0	1523.0	1714.84	0.00	TEMPERATURE (°C) SPAN (mV)		
1214.84	659122.0	1523.0	1214.84	0.00			
614.88	592404.0	1523.0	614.89	0.00			
14.66	525799.0	1524.0	14.67	0.00			
					5.50	311845.82	
					25.70	314071.83	
					34.05	314236.32	

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

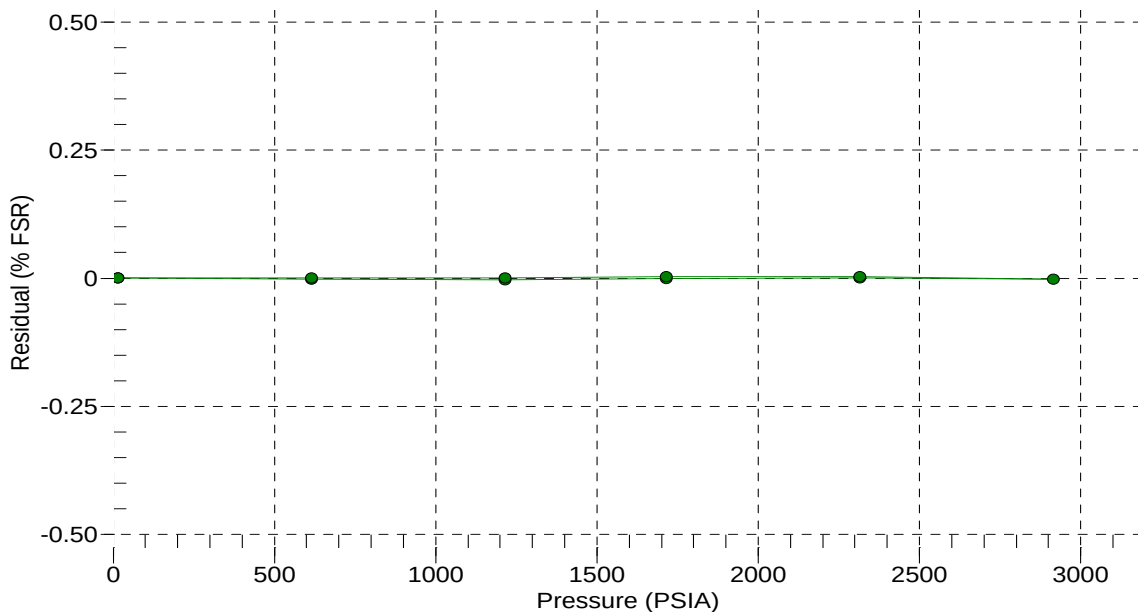
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 25-Sep-17 -0.00





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Pressure Test Certificate

Test Date: **2017-10-09**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16239**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **TH**

